

Proposal: Defending Against LLM Hallucinations using Lie Group Theory

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1 Abstract

We wish to research the process of creating and use-case testing various LLM prompts, such as prompting the user for additional details relevant to their project, assisting them in resummarising and elaborating on the information stored within a workspace, and generating intelligence from a collection of notes.

1.1 Background

There exist various strategies for taking notes, ranging from disciplines such as Zettelkasten, which focus on the workflow, to Markdown, which focus on the format.

Additionally, various cloud-based and desktop-based tools exist to facilitate the note taking in a canvas-based approach, where the notes are stored alongside spatial information.

Layout and format of the notes, paired with markup is often necessary in order to make an approach of looking into the information written down effective, yet no commercially-available process markets itself as primarily-oriented around the process of writing down the information in as much detail as possible.

1.2 Objectives

Our objective is to create and document a framework for prompt engineering that could be used as part of a note-taking process, in collaborative or solo setting, which aids the user by providing a toolbox of atoms of interaction with the LLM.

These atoms will be defined by tools and parsers that can be applied to any commercially available off-the-shelf LLM, in hopes of producing a useful toolbox for notetaking, creative writing, documentation writing, etc.

Keywords: large language models, notetaking, prompt engineering, collaborative editing

2 Introduction

Effective software design often relies on iterative and incremental refinement of both code and text-based documentation. However, traditional workflows can fragment this process, requiring constant context switching between development environments, notetaking applications, and external tools. A novel way to take notes with the help of LLMs (Large Language Models) is proposed, and we wish to make the process of interacting with the AI as deterministic as possible. Through a process of procedural prompt constructions, we aim to create an application that is able to assist the user in a predictable manner, harnessing the power of text transformers.

2.1 Aims

We aim to produce a library of datastructures and functions that can be composed together in order to ergonomically and optimistically produce a library of prepared messages for the LLM worker, that produce consistent and useful results.

We must create a testing methodology, and benchmark its efficacy, across multiple locally available LLMs.

3 Motivation

Various tools exist in the space of note-taking, note-organisation, documentation-writing. Few of these tools have managed to successfully leverage the LLM economy from the point of collaborative editing with aidance from the LLMs.

Our motivation for this project comes directly from the point of view that a creative block, writers paralysis, demotivation and low mood which can directly affect the cadence of a written project, but can be tackled with the power of a carefully engineered software.

This project aims to fill this gap by developing a suite of tools that can be integrated into, or be used to build a note-taking application from the ground up, or to serve as a framework for specifying a toolkit in future work.